

* Majorana spinor Ψ in $D=10, 11$ has $32 = 2^{10/2}$ components

$D=11$: Only one super theory whose susy alg. contains a super charge which is Maj.

$$\Psi^M = (\Psi^M)^c$$

$D=10$: Ψ : Majoran-Weyl : 16-components

• Two maximal supergravity theories

(i) IIA: $([1001] \oplus [1010]) \otimes ([1001] \oplus [?])$

2: Maj.

Two-Maj. Weyl. of opposite chirality

(ii) IIB: Two: Maj. Weyl. of same chirality

↓ (truncate)

(iii) Type I: Single Maj.-Weyl. : 16-components

→ couples to 10d Super Yang Mills

* Super in 10d/11d are uniquely determined by their susy and that they possess no more than two spacetime derivatives

10d
→ Super are the complete low energy eff. actions for IIA and IIB superstring theories

→ led us to realize that branes must be included

Four ways to Construct super theories:

(i) Noether method

(ii) Super Space

→ $T \Sigma^{(n,m)}$ acts on all objects the same

→ Problem amounts to finding which of the torsions and curvatures should be zero.

(iii) Gauging space-time groups

(iv) Dimensional Reduction

Noether Method:

Construct Yang-Mills from the linearised (free) theory: linearised theory of a spin-1 field

• Symmetries:

(i) Global trans.

$$e^{i\theta}$$

(ii) local abelian trans. $e^{i\theta(x)}$

(i) Belong to a Group S with generators R_i s.t.

$$[R_i, R_j] = S_{ij}^k R_k$$

The vector fields, $\Omega'(M) \otimes S$

$$A = A_a^i R_i dx^a$$

$$\Rightarrow \delta A = (e^{T R_j} A_a)^i = A_a^i + \delta A_a^i$$

Take A in the adjoint rep. then

$$(R_j)^l_k = S_{jk}^l, |T^i| \ll 1$$

$$\therefore \delta A_a^i = S_{jk}^i T^j A_a^k$$

(ii) local abelian

$$\delta A_a^i = \partial_a \Delta^i$$

$$A \rightarrow A_a^i + \partial_a \Delta^i$$

$$\Rightarrow \delta A_a^i = S_{jk}^i T^j \Delta^k$$

Form a closed algebra:

$$[\delta_\Delta, \delta_T] A_a^i = \partial_a (S_{jk}^i T^j \Delta^k)$$

The Action: $\int d^4x \mathcal{L}(A_a^i, \partial_a A_a^i)$

$$\delta_\Delta S = 0 = \int d^4x \frac{\delta \mathcal{L}}{\delta A_a^i} \delta_\Delta A_a^i + \frac{\delta \mathcal{L}}{\delta (\partial_a A_a^i)} \delta_\Delta (\partial_a A_a^i)$$

$$\Rightarrow \frac{\delta \mathcal{L}}{\delta A_a^i} \partial_a \Delta^i + \frac{\delta \mathcal{L}}{\delta (\partial_b A_a^i)} \partial_b (\partial_a \Delta^i)$$

$$\Rightarrow \frac{\delta \mathcal{L}}{\delta A_a^i} - \partial_b \frac{\delta \mathcal{L}}{\delta (\partial_b A_a^i)} = 0$$

$$g(0,0) + A_a^i \frac{\delta}{\delta A_a^i} \mathcal{L}(A_a^i, \partial_a A_a^i)$$

$$+ \frac{1}{2} A_a^i \frac{\partial}{\partial A_a^i} g(A_a^i, \partial A_a^i) \frac{\delta^2}{\delta A_a^i \delta A_b^j} g + \partial A_a^i \frac{\partial}{\partial A_a^i} g + \partial A_a^i \partial A_b^j \frac{\delta^2}{\delta A_a^i \delta A_b^j} g$$

$$+ \mathcal{O}(\partial^3)$$

I think this is right... went and found a term times $\frac{\delta^2 g}{\delta(\partial_a A_a^i) \delta(\partial_a A_b^j)}$

which goes as $\sim S_{ij} g^{ab} g^{cd}$, S_{ij} : Cartan-Killing

$$S^{(0)} = \int d^4x \{ -f_{ab}^i f^{jab} \} S_{ij}$$

$\leadsto$ Make global T^i local: $T^i(x)$

$\leadsto S^{(0)}$ is no longer invariant under

$$\delta A_a^i = S_{ij}^i T^j(x) A_a^j$$

$$\Rightarrow \delta S^{(0)} = \int d^4x \{ (\partial_a T^i(x)) j^{aj} \} S_{ij}$$

$$j^{ak} = -S_{ij}^k A_b^i f^{abj} \quad (\text{Noether Current})$$

$$\partial_a j^{ak} = 0$$

* Next: Consider

$$S_2 = S^{(0)} - \frac{1}{2} g \int d^4x (A_a^i j^{aj}) S_{ij}$$

g : gauge coupling constant

$\leadsto$ invariant to order g^0 provided we combine local trans. $T^i(x)$ w local trans. $\Delta^i(x)$

$$\Delta^i(x) = \frac{1}{g} T^i(x)$$

separate

linearised

The local and global trans. of the theory become knitted together into a single trans.

$$\delta = \delta_\Delta + \delta_T$$

$$\delta_T A_a^i = S_{ij}^i T^j(x) A_a^j$$

$$\delta_\Delta A_a^i = \Delta^i(x) A_a^i$$

$$\delta A_a^i = \frac{1}{g} \partial_\mu T^i(x) + S_{ij}^i T^j(x) A_a^k$$

$$\Rightarrow \delta A_1 = \delta A^{(0)} - \delta \left(\frac{1}{2} g \int d^4x (A_a^i j^{aj}) \right) S_{ij}$$

$$+ \frac{g}{2} \int d^4x \delta (A_a^i S_{ik}^j A_b^l (f^{bak})^k) S_{ij}$$

$$g \frac{1}{2} \int d^4x S_{ik}^j S_{ij} \left(\delta A_a^i A_b^l f^{bak} + A_a^i \delta A_b^l f^{bak} + A_a^i A_b^l (\delta f^{bak})^k \right)$$

nd Find terms of $\mathcal{O}(g)$, all others of $\mathcal{O}(1)$ will cancel with $\delta A^{(0)}$

$$= g/2 \int d^4x S_{ik}^j S_{ij} \left[(S_{nm}^i T^n(x) A_a^m) A_b^l (f^{ab})^k + A_a^i S_{nm}^l T^n(x) A_b^m f^{bak} + A_a^i A_b^l \delta f^{abk} \right]$$

$$\begin{aligned} \Rightarrow \delta f^{abk} &= \partial^a \delta A^{bk} - \partial^b \delta A^{ak} \\ &= \partial^a (S_{nm}^k T^n(x) A^{mb}) - \partial^b (S_{nm}^k T^n(x) A^{ma}) \\ &= S_{nm}^k \left[\partial^a T^n(x) A^{mb} - \partial^b T^n(x) A^{ma} + T^n(x) (\partial^a A^{mb} - \partial^b A^{ma}) \right] \\ &= S_{nm}^k (\partial^a T^n(x) A^{mb} - \partial^b T^n(x) A^{ma}) \\ &\quad + S_{nm}^k T^n(x) (f^{ab})^m \end{aligned}$$

$$= g/2 \int d^4x S_{ik}^j S_{ij} \left[(S_{nm}^i T^n(x) A_a^m) A_b^l (f^{ab})^k + A_a^i S_{nm}^l T^n(x) A_b^m f^{bak} + A_a^i A_b^l \delta f^{abk} + A_a^i A_b^l S_{nm}^k (\partial^a T^n(x) A^{mb} - \partial^b T^n(x) A^{ma}) \right]$$

nd Some how only last term survives

$$= g/2 \int d^4x (S_{ik}^j S_{ij} A_a^i A_b^l S_{nm}^k \partial^a T^n(x) A^{mb} - S_{ik}^j S_{ij} A_a^i A_b^l S_{nm}^k \partial^b T^n(x) A^{ma})$$

$$\begin{aligned}
(\text{const.}) &: -S_{ikl} A_a^i A_b^l S_{nm}^k \partial^a T^n A^{mb} \\
&= -S_{lik} A_b^l A_a^i S_{nm}^k \partial^a T^n A^{mb} \\
&= S_{ilk} A_b^l A_a^i S_{nm}^k \partial^a T^n A^{mb} \\
&= S_{ilk} \delta_{ij} A_b^l A_a^i S_{nm}^k \partial^a T^n A^{mb}
\end{aligned}$$

$$\begin{aligned}
\delta S_1 &= g \int d^4x S_{ij} S_{ilk} A_a^i A_b^l S_{nm}^k (\partial^a T^n) A^{mb} \\
&= g \int d^4x S_{kij} A_a^i \dots \\
&= g \int d^4x S_{kn} S_{ij}^n A_a^i A_b^j S_{lm}^k \partial^a T^m A^{lb} \\
&= \int d^4x -g S_{kn} S_{ij}^n A_a^i A_b^j S_{lm}^k \partial^a T^m A^{lb} \\
&\leadsto \text{Sign comes from } S_{ml}^k \rightarrow S_{lm}^k
\end{aligned}$$

$$\partial^a T^m \rightarrow A^{am}$$

• Add this term to $A_{(1)}$

$$\begin{aligned}
A_{(2)} &= A_1 + \int d^4x \frac{g^2}{4} (A_a^i A_b^j S_{ij}^l) (A^{bl} A^{cm} S_{lm}^n) S_{kn} \\
&= \frac{1}{4} \int d^4x F_{ab}^k (F^{ab})^n S_{kn}
\end{aligned}$$

$$F_{ab}^i = \partial_a A_b^i - \partial_b A_a^i - g S_{ijk}^l A_a^j A_b^k$$

• A_2 is invariant under δ to all orders

$$[\delta_{T_1}, \delta_{T_2}] A_a^i = (\text{closed})$$

$\leadsto$ forms an algebra

Now look at $N=1, D=4$ Super

$N=1$ Graviton multiplet $(\frac{3}{2}, 2) \oplus_{\text{CPT}} (-2, -\frac{3}{2})$

• Linearized theory has rigid susy and local

• abelian gauge inv.

• Take massless case:

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2 P_{\alpha\dot{\beta}}$$

$$P_\mu = (E, 0, 0, E)$$

$$\sigma^\mu p_\mu = \sigma^0 p_0 + \sigma^3 p_3 = E (\sigma^0 + \sigma^3) \\ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\therefore \{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 4E \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

all zero except for $\{Q_1, \bar{Q}_{\dot{1}}\} = 4E$

$$a \equiv \frac{1}{\sqrt{4E}} Q_1, \quad a^\dagger \equiv \frac{1}{\sqrt{4E}} \bar{Q}_{\dot{1}} \quad | \lambda_0 : \text{helicity}$$

$$| \lambda_0 \rangle \rightarrow a | \lambda_0 \rangle = 0 \quad | \lambda_0 \rangle : \text{vacuum}$$

$| \lambda_0 \rangle$: Carries some irrep of Poincare'

$$a^\dagger | \lambda_0 \rangle = | \lambda_0 + 1/2 \rangle, \quad \lambda_0 = 3/2 : \psi_\mu^\alpha$$

$$\bullet \gamma_5 = -i \gamma^0 \gamma^1 \gamma^2 \gamma^3$$

$$1 = 2 : \psi_{\mu\nu}$$

$$(\psi_{\mu\nu}, \psi_{\mu\alpha})$$

$\psi_{\mu\nu}$: Symmetric and traceless

$$\eta^{\mu\nu} \psi_{\mu\nu} = 0$$

$$\leadsto g_{\mu\nu} = \eta_{\mu\nu} + \psi_{\mu\nu}$$

$\bullet$ inv. under abelian diffeomorphisms:

$$\delta \psi_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

$$\delta R_{\mu\nu}^L = 0$$

$$X^\mu \rightarrow x^\mu + \xi^\mu$$

$$\bullet \text{ de Donder gauge: } \partial^\rho \psi_{\rho\nu} - \partial_\nu \psi^\rho{}_\rho / 2 = 0 \\ \text{or } \partial^\rho \psi_{\rho\nu} = 0 \Rightarrow \partial^2 \psi_{\mu\nu} = 0$$

$$\text{Sugra fields: } E_{\mu\nu} = 0 \text{ and } R^\mu{}_\mu = 0 \\ E_{\mu\nu} = R_{\mu\nu}^L - \frac{1}{2} \eta_{\mu\nu} R^L = 0$$

$$R^\mu{}_\mu = \epsilon^{\mu\nu\rho\kappa} i \gamma_5 \gamma_\nu \partial_\rho \psi_\kappa = 0$$

$$\hookrightarrow \delta \psi_{\mu\alpha} = \partial_\mu \theta_\alpha(x)$$

$\psi_{\mu\alpha}$: Majorana Vector

$$\Rightarrow \text{Dirac } \delta \psi_\mu = 0 = \gamma^\mu \psi_\mu = \partial^\mu \psi_\mu$$

Symmetry: $\delta h_{\mu\nu} = [\xi Q + \bar{\xi} \bar{Q}, h_{\mu\nu}]$

$$= \xi^\alpha [Q_\alpha, h_{\mu\nu}] + \bar{\xi}_{\dot{\alpha}} [\bar{Q}^{\dot{\alpha}}, h_{\mu\nu}]$$

$$= \frac{i}{2} \bar{\xi} (\gamma_\mu \psi_\nu + \gamma_\nu \psi_\mu) + \delta_{\mu\nu} \gamma^\kappa \psi_\kappa$$

$$Q^\dagger : |2\rangle \rightarrow |3/2\rangle$$

$$Q : |2\rangle \rightarrow |3/2\rangle \leftarrow \underline{\text{zero}}$$

so only $Q^\dagger$
are likewise for

$$\delta \psi_\mu = + \delta_2 \sigma^{ab} \partial_a h_{b\mu} \xi + \delta_3 \partial_\nu h^\nu_\mu \xi$$

$$= [\xi Q, \psi_\mu]$$

$$\Rightarrow \delta \psi_\mu \Rightarrow \psi_\mu(x) \rightarrow \psi_\mu(x) + \partial_\mu \mathcal{H}(x)$$

• The alg. should close:

$$[\delta_\eta, \delta_\xi] h_{\mu\nu} = \delta_\eta \left(\frac{i}{2} \bar{\xi} \gamma_\mu \psi_\nu + \bar{\xi} \gamma_\nu \psi_\mu \right) + \delta_\eta (\delta_{\mu\nu} \bar{\xi} \gamma^\kappa \psi_\kappa)$$

since $\delta_\eta h_{\mu\nu} = 0$

$$= \frac{i}{2} [\bar{\xi} \gamma_\mu \partial_\nu \mathcal{H} + \bar{\xi} \gamma_\nu \partial_\mu \mathcal{H}] + \delta_{\mu\nu} \bar{\xi} \gamma^\kappa \partial_\kappa \mathcal{H}$$

for $\rho_\mu \equiv \frac{i}{2} \bar{\xi} \gamma_\mu \mathcal{H}$

$$[\delta_\eta, \delta_\xi] h_{\mu\nu} = \partial_\mu \rho_\nu + \partial_\nu \rho_\mu + 2 \delta_{\mu\nu} \partial_\kappa \rho^\kappa$$

so this closes on diffeomorphisms of $h_{\mu\nu}$

provided $\delta_\nu = 0$

$$\Rightarrow [\delta_\xi, \delta_\xi] h_{\mu\nu} = 0$$

as expected since $[P, Q] = 0$

$$[\delta_\xi, \delta_\xi] \psi_\mu = \delta_\xi \delta_\xi \psi_\mu \quad \text{since } \delta_\xi \psi_\mu = 0$$

$$\delta_\xi \left(\delta_2 \sigma^{ab} \partial_a h_{b\mu} \xi + \delta_3 \partial_\nu h^\nu_\mu \xi \right)$$

$$= \delta_2 \sigma^{ab} \partial_a (\partial_b \xi_\mu + \partial_\mu \xi_b) + \delta_3 \partial^\nu (\partial_\nu \xi_\mu + \partial_\mu \xi_\nu) \xi$$

$\Rightarrow \sigma^{ab} = -\sigma^{ba} \Rightarrow$ first term vanishes

$$\hookrightarrow = \delta_2 \sigma^{ab} \partial_a (\partial_m \xi_b) \xi + \delta_3 \partial_r \partial^r \xi \xi + \delta_3 \partial_r \partial_m \xi^m \xi$$

can be a local transf. if

$$Q = \delta_2 \sigma^{ab} \partial_a \xi_b \xi$$

and $\delta_3 = 0$

$$\therefore \boxed{\delta_1 = \delta_3 = 0}$$

$$\Rightarrow \delta_a h_{\mu\nu} = \frac{1}{2} (\bar{\xi} \gamma_\mu \psi_\nu + \bar{\xi} \gamma_\nu \psi_\mu)$$

$$\delta_\alpha \psi_\mu = \delta_2 \sigma^{ab} \partial_b h_{b\mu} \xi$$

* Now test comm. of two susy

$$\begin{aligned} [\delta_{\xi_1}, \delta_{\xi_2}] h_{\mu\nu} &= \delta_{\xi_1} \left(\frac{1}{2} [\bar{\xi}_2 \gamma_\mu \psi_\nu + \bar{\xi}_2 \gamma_\nu \psi_\mu] \right) \Leftrightarrow \\ &= \delta_2 \frac{1}{2} \bar{\xi}_2 \gamma_\mu (\sigma^{ab} \partial_a h_{b\nu} \xi_1) + \delta_2 \frac{1}{2} \bar{\xi}_2 \gamma_\nu (\sigma^{ab} \partial_b h_{b\mu} \xi_1) \\ &\quad - (1 \leftrightarrow 2) \end{aligned}$$

$$\leadsto \bar{\xi}_2 \gamma_\mu \left(\frac{1}{2} (\sigma^a \bar{\sigma}^b - \sigma^b \bar{\sigma}^a) \right)$$

• Note $\bar{\sigma}^m \sigma^r \bar{\sigma}^s \sim R^{ma} R^{rs} + \dots$

* Linear Action:

$$\boxed{S_0 = \int d^4x \left(-\frac{1}{2} h^{\mu\nu} E_{\mu\nu} - \frac{1}{2} \bar{\psi}_\mu R^\mu \right)}$$

w $\delta_1 = \delta_3 = 0, \delta_2 = -1$

• Now find Auxiliary fields..

$\leadsto$ Final theory:

$$\boxed{S = \int d^4x \left\{ \frac{e}{2\kappa^2} R - \frac{1}{2} \bar{\psi}_\mu R^\mu - \frac{1}{3} e (M^2 + N^2 - b_\mu b^\mu) \right\}}$$

$$R = R_{\mu\nu}{}^{ab} e_a^\mu e_b^\nu, \quad R_{\mu\nu}{}^{ab} \frac{\sigma_{ab}}{4} = [D_\mu, D_\nu]$$

$$R^\mu = e^{\mu\nu\rho\sigma} i \gamma_5 \gamma_\nu D_\rho \psi_\sigma$$

or using

$$M = N = b_\mu = 0$$

$$S_{\text{on-shell}} = \int d^4x \left\{ \frac{c}{2u} R - \frac{1}{2} \psi_\mu \epsilon^{\mu\nu\rho\sigma} \gamma_5 \gamma_\nu D_\rho \psi_\sigma \right\}$$

On-shell SuperSpace:

- find dim of possible gauge inv. quantities of the theory
- Compare with dim of superspace tensors & covectors to find the latter have dim which are lower than any of the gauge inv. quantities

* $D=4, N=1$ super space $(x^\mu, \theta^\alpha) = \mathbb{Z}^\pi$
 $\Rightarrow$ Supergeneral coordinate rep.
 $\mathbb{Z}^\pi \rightarrow \mathbb{Z}^{\pi_1} = \mathbb{Z}^\pi + \mathbb{Z}_g^\pi$

where

$$\mathbb{Z}_g^\pi = (\mathbb{Z}^\mu, \mathbb{Z}^\alpha)$$

* Scalar superfields: $\Phi(z') = \Phi(z)$

$$\psi_\Lambda = \frac{\partial \Phi}{\partial z^\Lambda} \quad \Lambda: \text{superspace index}$$

$$z^\mu \sim (z^\mu, \theta^\alpha, \bar{\theta}_{\dot{\alpha}})$$

$$z^\mu z^\nu = (-1)^{\mu\nu} z^\nu z^\mu$$

* Exterior product: $dz^\mu \wedge dz^\nu = (-1)^{\mu\nu} dz^\nu \wedge dz^\mu$
 $dz^\mu \cdot z^\nu = (-1)^{\mu\nu} z^\nu dz^\mu$

where the last term is $dz^\mu \wedge z^\nu$

• Super-p-form

$$\Omega^{(p)} = dz^{\mu_1} \wedge \dots \wedge dz^{\mu_p} W_{\mu_1 \dots \mu_p}(z)$$

• Functions of z are 0-forms:

$$F(z)$$

1-form

$$\Delta = dz^\mu W_\mu(z) = dx^\mu W_\mu(z) + d\theta^\alpha W_\alpha(z) + d\bar{\theta}_{\dot{\alpha}} W^{\dot{\alpha}}(z)$$

Exterior derivative $dF = dz^\mu \frac{\partial F}{\partial z^\mu}$ (0-form)

$\leadsto$ For a p -form $Q = dz^{\mu_1} \wedge \dots \wedge dz^{\mu_p} W_{\mu_1 \dots \mu_p}(z)$
 $dQ = dz^{\mu_1} \wedge \dots \wedge dz^{\mu_p} \wedge dz^N \partial_N W_{\mu_1 \dots \mu_p}(z)$
 $d^2 = 0$

* Super differential form are covariant under coordinate changes.

$\leadsto$ Pull back: $\phi: M \rightarrow M$ (diffeo)

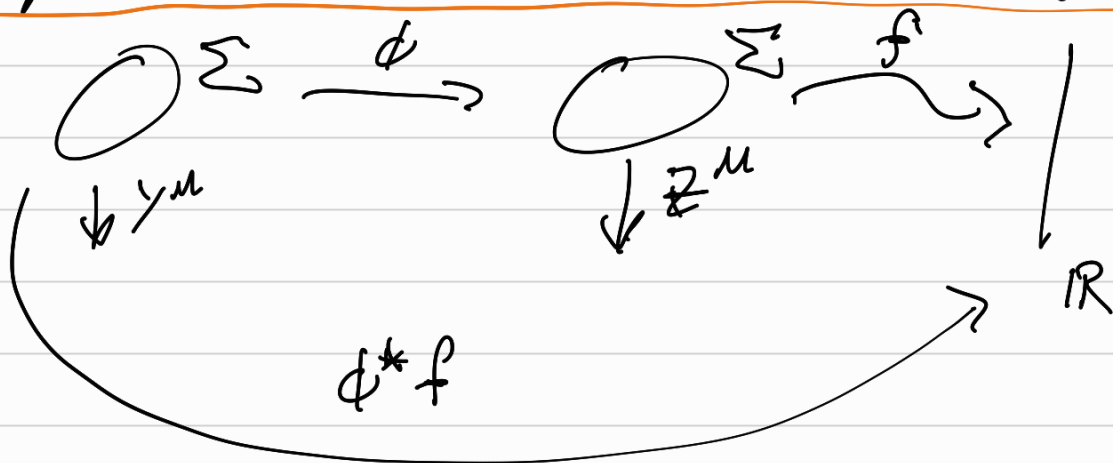
$$f: M \rightarrow \mathbb{R}$$

$$\phi^* f = f \circ \phi$$

$$f = \gamma, \quad f \circ \phi = \gamma(z)$$

$$= F'(z')$$

$$\boxed{F(y^\mu) = F(y^\mu(z)) = \phi^* F(z^\mu)}$$



* Supervielbeins: Vielbein 1-form

$$EA = dz^M E_M^A(z)$$

$$\phi^* E_M^A(z') = E_M'^A(z') = \frac{\partial z^N}{\partial z'^M} E_N^A(z)$$

and so for infinitesimal diffeos:

$$z^{\mu'} = z^\mu + \xi^\mu(z)$$

$$E'^A_M(z) = \xi^\mu(z) \partial_\mu E_N^A(z) = \delta^N_M E_N^A(z) + \partial_\mu \xi^\mu E_N^A(z)$$

$$\boxed{\delta E_\pi^N = \xi^\Lambda \partial_\Lambda E_\pi^N + (\partial_\pi \xi^\Lambda) E_\Lambda^N}$$

N, M : tangent space indices (Flat)

π, Λ : world indices (Curved Spacetime)

$$e_m^M : m : MD \quad \eta_{mn}$$

$$\mu : MD \quad g_{\mu\nu}$$

* Tangent space Group : Lorentz $SO(1,3)^+$

$$\delta E^A = \delta (E_\Lambda^A dz^\Lambda) = \delta E_\Lambda^A dz^\Lambda$$

• only acts on A (flat) indices

$$\delta E_\pi^N = E_\pi^M \Delta_M^N$$

$$\Delta_M^N = \lambda_n^m \oplus -\frac{1}{4}(\sigma_{mn})_A^B \Delta^{mn} \oplus +\frac{i}{4}(\sigma_{mn})_{\dot{A}}^{\dot{B}} \Delta^{mn}$$

$$\Delta_M^N = \begin{pmatrix} \lambda_n^m & 0 & 0 \\ 0 & -\frac{1}{4}(\sigma_{mn})_A^B \Delta^{mn} & 0 \\ 0 & 0 & +\frac{i}{4}(\sigma_{mn})_{\dot{A}}^{\dot{B}} \Delta^{mn} \end{pmatrix}$$

$\Delta^{mn} : f^n$ on superspace rotations of vec. & group

• Σ^g : Could choose a larger tangent space group

Δ_M^N could be an isometry of the theory in Σ^g

$\leadsto$ Take the metric

$$g_{NM} = \begin{pmatrix} a_1 \eta_{mn} & 0 & 0 \\ 0 & a_2 \xi_{\alpha\beta} & 0 \\ 0 & 0 & a_3 \xi_{ij} \end{pmatrix}$$

• Since $g_{NM} = g_{NM}^*$, $a_1^* = a_1$
 but $a_2^* \xi_{\alpha\beta} = a_3 \xi_{\alpha\beta}$ $\Rightarrow$ $a_2^* = a_3$

and so we may scale away a_2 . This takes tangent space group to be

$$OSp(4,1)$$

Take $g_{\pi\Delta} = E_{\pi}^N g_{NM} E_{\Delta}^M$ and this will mimic Relativity. However this will NOT lead to the x-space component of $N=1$ sugra.

$OSp(4,1)$ yields a higher derivative action.
Tangent Supervectors

$$V^N = V^{\pi} E_{\pi}^N$$

transform as irreps of Lorentz.

$V^m V_m, V^{\alpha} V^{\beta} \epsilon_{\alpha\beta}, V^{\dot{\alpha}} V^{\dot{\beta}} \epsilon_{\dot{\alpha}\dot{\beta}}$
are all separately invariant.

* Define Lorentz valued Spin Connection

$$(\Omega_{\Delta})_N = \begin{pmatrix} (\Omega_{\Delta})_m^N & 0 & 0 \\ 0 & -\frac{1}{4} \Omega_{\Delta}^{mn} (\sigma_{mn})_A^B & 0 \\ 0 & 0 & \frac{1}{4} \Omega_{\Delta}^{mn} (\bar{\sigma}_{mn})_{\dot{A}}^{\dot{B}} \end{pmatrix}$$

* Transforms under Supergroupal coord. transf.

$$\delta(\Omega_{\Delta})_N = \left[\mathcal{L}_{\xi} \Omega_{\Delta}^N \right]_{\Delta} = \xi^{\pi} \partial_{\pi} \Omega_{\Delta}^N + \partial_{\Delta} \xi^{\pi} \Omega_{\pi N}^N$$

$$(z^m)' = z^m - \xi^m(z)$$

* Under Lorentz:

$$\delta \Omega_{\Delta}^N = -\partial_{\Delta} \Omega_m^N + \Omega_{\Delta m}^S \Lambda_S^N + \Omega_{\Delta R}^N \Lambda_R^M (-1)^{(M+R)(N+R)}$$

Aside: For gauge we have 2 valued

p-forms:

$$\leadsto A = A_{\pi} dz^{\pi} = A_{\pi(z)}^m T_m(z) dz^{\pi}$$

Thus, under an exterior der. of $A^{(p)}$

$$\begin{aligned} dA &= dz^{\mu_1} \wedge \dots \wedge dz^{\mu_p} \wedge dz^N \left(\frac{\partial}{\partial z^N} A_{\mu_p \dots \mu_1}(z) \right) \\ &= dz^{\mu_1} \wedge \dots \wedge dz^{\mu_p} \wedge dz^N \left(\frac{\partial}{\partial z^N} A_{\mu_p \dots \mu_1}^M \right) T_M + A^L \frac{\partial}{\partial z^N} T_L \end{aligned}$$

so we have

$$dA = dA^L \otimes T_L + A^L \wedge dT_L$$

$\Rightarrow$ Want to account for dT_L term: Introduce a connection Ω :

Connection: L.A. valued 1-form

$$A = dz^{\pi} A_{\pi}^N (i T_N)$$

which has the transformation law:

$$A' = X^{-1} A X + X^{-1} dX$$

Def. the covariant derivative

$$\mathcal{D}\Omega = d\Omega + \Omega \wedge A$$

$$\mathcal{D}\Omega = dz^N \mathcal{D}_N \Omega$$

$$= dz^{\pi_1} \dots dz^{\mu_p} dz^N \frac{\partial}{\partial z^N} \Omega_{\mu_p \dots \mu_1}$$

$$+ dz^{\pi_1} \dots dz^{\pi_p} dz^N \frac{\partial}{\partial z^{\pi}} i A_{\pi}^M T_M \Omega_{\mu_p \dots \mu_1}$$

* Curvature: $F = dA + A \wedge A$

$$D_\pi = e_\pi^N \frac{\partial}{\partial z^N} \quad \Big| \quad dz^M \frac{\partial}{\partial z^N} = \delta^M_N \quad D_\pi$$

$$D_\Lambda = \partial_\Lambda + \frac{1}{2} \Omega_\Lambda^{mn} J_{mn}$$

J_{mn} : Lorentz
Generators

In Lorentz: $D_M = E_M^\Lambda (\partial_\Lambda + \frac{1}{2} \Omega_\Lambda^{mn} J_{mn})$

$$E_N^\Lambda E_\Lambda^M = \delta_N^M \quad \Big| \quad E_\Lambda^M E_M^\pi = \delta_\Lambda^\pi$$

$$D_M = (D_M, D_\alpha, D_{\bar{\alpha}})$$

$$D_M = \frac{\partial}{\partial x^M} \equiv \partial_M$$

$$D_\alpha = \partial / \partial \theta^\alpha - i(\sigma^m)_{\alpha\dot{\alpha}} \bar{\Theta}^{\dot{\alpha}} \partial_m$$

$$D_{\dot{\alpha}} = \partial / \partial \bar{\Theta}^{\dot{\alpha}} - i(\sigma^m)^{\alpha\dot{\alpha}} \Theta^\alpha \partial_m$$

Torsion and Curvature:

$$[D_M, D_N] = T_{MN}^R D_R + \frac{1}{2} R_{NM}^{mn} J_{mn}$$

implies: $R_{MN}{}^{rs} = 0$

• Torsion: Covariant der. of the vielbein

$$\begin{aligned} T^A &= dE^A + E^B A_B^A = \frac{1}{2} dz^M dz^N T_{NM}^A \\ &= \frac{1}{2} E^C E^B T_{BC}^A \end{aligned}$$

Curvature: $R = d\Omega + \Omega \wedge \Omega$
Here $A^{(1)} = \Omega^{(1)}$

$$T_{MN}^R = e_M^\Lambda \partial_\Lambda E_N^\pi E_\pi + \Omega_{MN}^R - (-1)^{MN} (M \leftrightarrow N)$$

$$K_{MN} = E_M^\mu E_N^\nu (-1)^{\epsilon(M)\epsilon(N)} \{ \partial_\Delta \Omega_\mu^\nu + \Omega_\Delta^\mu \Omega_{\pi\mu}^\nu - (-1)^{\Delta\pi} (\Delta\Delta \rightarrow \pi) \}$$

Convention: $(\frac{1}{2} J_{pq} \omega^{pq}) V_n = \omega_n^m V_m$

$$\{D_N, D_M\} = [e_N^\Delta (\partial_\Delta + \frac{1}{2} \Omega_\Delta^{\mu\nu} J_{\mu\nu}), e_M^\pi] \\ e_N^\Delta (\partial_\Delta e_M^\pi) \partial_\pi + e_N^\Delta e_M^\pi \partial_\Delta \partial_\pi - (-1)^{\mu\nu} \partial_\pi e_N^\Delta e_M^\pi \partial_\Delta \\ - e_N^\Delta e_M^\pi \partial_\Delta \partial_\pi$$

$$= e_N^\Delta (\partial_\Delta e_M^\pi) \partial_\pi + (-1)^{\mu\nu} e_M^\pi \partial_\pi (e_N^\Delta) \partial_\Delta \quad \textcircled{\star}$$

• (ii) $[e_N^\Delta \partial_\Delta, \frac{1}{2} e_M^\pi \Omega_\pi^{\mu\nu} J_{\mu\nu}]$

$$e_N^\Delta \left[\partial_\Delta e_M^\pi \Omega_\pi^{\mu\nu} J_{\mu\nu} + (-1)^{\pi\Delta} e_M^\pi \partial_\Delta \Omega_\pi^{\mu\nu} J_{\mu\nu} \right. \\ \left. + e_M^\pi \Omega_\pi^{\mu\nu} J_{\mu\nu} \partial_\Delta \right] \\ = (-1)^{\mu\nu} (e_M^\pi \Omega_\pi^{\mu\nu} J_{\mu\nu}) e_N^\Delta \partial_\Delta$$

$$= \frac{1}{2} e_N^\Delta \left(\partial_\Delta e_M^\pi \Omega_\pi^{\mu\nu} J_{\mu\nu} + (-1)^{\pi\Delta} e_M^\pi \partial_\Delta \Omega_\pi^{\mu\nu} J_{\mu\nu} \right)$$

$\Rightarrow \dots$

$$[A_M, B_N] = A_M B_N - (-1)^{\mu\nu} B_N A_M$$

* Supergeneral coordinate trans.

$$\delta E_\Delta^\mu = - E_\Delta^R \xi_N^N T_{NR}^\mu + D_\Delta \xi_N^\mu$$

$$\delta \Omega_{\Delta\mu}^N = E_\Delta^R \xi_N^P R_{PR\mu}^N$$

Bianchi: $[D_M, [D_N, D_R]] - [[D_M, D_N], D_R] + (-1)^{RN} \\ [[D_M, D_R], D_N] = 0$

* Via dimensional analysis: $[D_N, D_m] = \gamma_{NM}^R D_R + \frac{1}{2} R_{NM}{}^{rs} \gamma_{rs}$
 F : Fermionic } (index)
 B : Bosonic

$$[D_B] = 1, [D_F] = 1/2$$

$$\therefore [T_{FF}^B] = 0, [T_{FF}^F] = [T_{FB}^B] = 1/2$$

$$[T_{FB}^F] = [T_{BB}^B] = 1$$

$$[T_{BB}^F] = 3/2$$

• Note: this can be seen as adding the bottom two and subtracting the top index:

$$T_{FB}^F \rightsquigarrow (\frac{1}{2} + 1) - 1/2 = 1$$

$$T_{FF}^B = (\frac{1}{2} + 1/2) - 1 = 0$$

etc...

$$[R_{FF}^{mn}] = 1, [R_{FB}^{mn}] = 3/2, [R_{BB}^{mn}] = 2$$

(Same idea)

* This implies: $[e_m^\mu] = 0, [\psi_m^\alpha] = 1/2, [M] = [N] = [b_m] = 1$

• On-shell states: h_{mn}, ψ_m^α

$$\delta h_{mn} = \partial_m \xi_n + \partial_n \xi_m, \delta \psi_m^\alpha = \partial_m \theta^\alpha$$

• Ignored non-linear terms: i.e. start with linearized super.

$$[h_{mn}] = 0, [\psi_m^\alpha] = 1/2 \text{ (Geometric dimension)}$$

• Highest/lowest dim gauge objects invariant under the above are given by

$$[\partial\psi] = 3/2$$

$$[\partial\partial h] = 2$$

$$* T_{MN}{}^R \sim x^m + \Theta + \bar{\Theta}^{\dot{\alpha}}$$

• These objects at $\theta = 0$ must correspond to

gauge invariant x-space objects.

If no such object exists, then the corresponding tensor vanishes at $\theta=0$ (and all other θ).

$$\left[\begin{array}{l} [T_{FF}^B] = 0 \Rightarrow T_{\alpha\beta}^m, T_{\dot{\alpha}\dot{\beta}}^m, T_{\alpha\dot{\beta}}^m, T_{\dot{\alpha}\beta}^m \end{array} \right.$$

• The only invariant object at $[T] = 0$ is $\sigma^m_{\alpha\dot{\alpha}}$

$$\Rightarrow T_{\alpha\beta}^m = T_{\dot{\alpha}\dot{\beta}}^m = 0$$

and

$$T_{\alpha\dot{\beta}}^m = c (\sigma^m)_{\alpha\dot{\beta}}$$

• Reality conditions of $T \Rightarrow c \in \mathbb{C}$. Normalise to $c = -2i$

$$T_{\alpha\dot{\beta}}^m = -2i (\sigma^m)_{\alpha\dot{\beta}}$$

$$[T] = 1/2 : [T_{FF}^F] = [T_{BF}^B]$$

• No such covariant objects

$$T_{\alpha\beta}^{\gamma} = T_{\alpha\dot{\beta}}^{\gamma} = T_{\dot{\alpha}\beta}^{\gamma} = T_{\alpha m}^n = 0$$

$$[T] = 1, [T_{BF}^F] = [T_{BB}^B], [R_{FF}^{mn}]$$

• No such invariant in x-space at $\theta=0$:

w_m^{rs} would be if it an independent object however if 1st order it is not independent

$$T_{n\alpha}^{\dot{B}} = T_{n\alpha}^B = R_{\alpha\beta}^{mn} = 0 = R_{\alpha\dot{\beta}}^{mn}$$

Note: only writing "half" of the expressions since the reality condition $(T_{\alpha\dot{\beta}})^* = T_{\dot{\alpha}\beta} = T_{\alpha\dot{\beta}}$

* Every dim: 0, 1/2, 1 variables aside from $T_{\alpha\dot{\beta}}^m = -2i (\sigma^m)_{\alpha\dot{\beta}}$

$$[T] = 3/2 : [T_{BB}^F], [R_{FB}^{mn}] = 3/2$$

• Note we have $[D\psi] = 3/2$

$$T_{mn}^\alpha, R_{\alpha\beta}^{mn}$$

so we can find the on-shell theory from Bianchi:

$$I_{nB\dot{D}}^{\dot{C}} = -D_n T_{B\dot{D}}^{\dot{C}} + T_{nB}^F T_{F\dot{D}}^{\dot{C}} + R_{nB\dot{D}}^{\dot{C}} + \dots$$

yields

$$-2i(\sigma^m)_{B\dot{D}} T_{mn}^{\dot{C}} - R_{nB\dot{D}}^{\dot{C}} = 0$$

so Trace over $\dot{D}^{\dot{C}}$

$$2i(\sigma^m)_{B\dot{D}} T_{mn}^{\dot{D}} + \underbrace{R_{nB\dot{D}}^{\dot{D}}}_{\text{Sym in last two but } \dot{D} \dot{D} \text{ is anti sym}} = 0$$

Sym in last two but $\dot{D} \dot{D}$ is anti sym

$\Rightarrow$ Rarita - Schwinger equation:

$$(\sigma^m)_{B\dot{D}} T_{mn}^{\dot{D}} = 0$$

$$[T] = 2 \mid R_{B\dot{B}}^{mn}$$

$$I_{Bmn}^\alpha = \dots$$

apply constraints:

$$-D_B T_{mn}^\alpha + R_{mnB}^\alpha = 0$$

• Contracting w $(\sigma^m)_{\alpha\dot{\beta}}$

$$(\sigma^m)_{\alpha\dot{\beta}} D_B T_{mn}^\alpha = R_{mnB}^\alpha (\sigma^m)_{\alpha\dot{\beta}} \approx 0$$

Note:

$$R_{mnB}^\alpha = -\frac{1}{4} R_{mn}^{PQ} (\sigma_{PQ})_B^\alpha$$

$$(RHS) -\frac{1}{4} R_{mn}^{PQ} (\sigma_{PQ})_B^\alpha (\sigma^m)_{\alpha\dot{\beta}}$$

$$\therefore R_{mn}^{PQ} (\sigma_{PQ} \sigma^m)_{B\dot{\beta}} = 0$$

$$\Rightarrow R_{mn} - \frac{1}{2} R_{mn} \eta = 0 \Rightarrow R_{mn} = 0$$

with $R_{mn} = R_{msn}{}^s$

$$E_m^N(\theta=0) = e_m^N, \quad E_m^A(\theta=0) = \frac{1}{2} \psi_m^A$$

$$\delta E_a^N(\theta=0) = \sum^\pi \partial_\pi E_a^N|_{\theta=0} + \partial_a \sum^\pi E_a^N|_{\theta=0}$$

$$= \dots + \partial_a \xi^m e_m^N + \dots$$

Choose $\partial_a \xi^m$ s.t. $E_a^N = 0$

Choose

$$E_{\dot{\alpha}}^{\dot{B}} = \delta_{\dot{\alpha}}^{\dot{B}}, \quad E_a^B = \delta_a^B, \quad E_{\dot{\alpha}}^{\dot{B}} = 0$$

Thus

$$E_\pi^M(\theta=0) = \begin{pmatrix} e_m^N & \frac{1}{2} \psi_m^A & \frac{1}{2} \psi_m^{\dot{A}} \\ 0 & \delta_a^B & 0 \\ 0 & 0 & \delta_{\dot{B}}^{\dot{\alpha}} \end{pmatrix}$$

$$\Omega_\mu^{mn}(\theta=0) \equiv \omega_\mu^{nm}$$

• Use Lorentz: $\Omega_\alpha^{mn}(\theta=0) \approx 0$

$$\delta \Omega_{\Delta\mu}^N = -\partial_\Delta \Omega_\mu^N + \Omega_{\Delta\mu}^S \Delta_S^N + \dots$$

i.e. $\delta \Omega_\alpha^{mn} = -\Omega_\alpha^{mn} \uparrow$ with a choice of Δ .

→ Evaluate $T_{\pi\Delta}^R = (-1)^{\pi(N+\Delta)} E_\Delta^N E_\pi^M T_{MN}^R$

$$= -\partial_\pi E_\Delta^R + \Omega_{\pi\Delta}^R - (-1)^{\pi\Delta} (\pi \leftrightarrow \Delta)$$

at $\theta=0$:

$$T_{mn}^{\dot{\alpha}} = -\frac{1}{2} \partial_m \psi_n^{\dot{\alpha}} + \Omega_{mn}^{\dot{\alpha}} - (m \leftrightarrow n)$$

• Note $\Omega_{mn}^{\dot{\alpha}} = E_n^N \omega_{mN}^{\dot{\alpha}} = \frac{1}{2} \psi_n^{\dot{B}} \omega_{m\dot{B}}^{\dot{\alpha}}$

$$T_{mn}^{\dot{\alpha}} = -\psi_{mn}^{\dot{\alpha}} \equiv D_m \psi_n^{\dot{\alpha}} = \partial_m \psi_n^{\dot{\alpha}} - \psi_n^{\dot{B}} \omega_{m\dot{B}}^{\dot{\alpha}}$$

$$\Rightarrow T_{mn}^{\dot{\alpha}} = e_m^N e_n^M T_{MN}^{\dot{\alpha}}:$$

$$0 = (\sigma^m)_{\alpha\dot{\beta}} T^{\dot{\beta}}_{mn}(\theta=0) = -\frac{1}{2} (\sigma^m)_{\alpha\dot{\beta}} e_m^N e_n^M \psi_{MN}^{\dot{\beta}}$$

$$\mathcal{O} = -\frac{1}{2} (\sigma^m)_{\alpha\dot{\beta}} e_m^\mu e_n^\nu \mathcal{D}_\mu \psi_{\dot{\alpha}} \dot{\beta} \quad (\text{Rarita-Schwinger})$$

• Similar for Spin 2 term:

The Action is given by

$$S = \frac{1}{2\kappa^2} \int d^8z E$$

$$E = \det(E_n^N)$$